Python-Matplotlib

Objective

- Basics of Matplotlib
- Installation
- Types Of Plots
- Summary

Basics of Matplotlib

What is Matplotlib?

1. Matplotlib is a low level graph plotting library in python that serves as a visualization utility.
2. Matplotlib was created by John D. Hunter.
3. Matplotlib is open source and we can use it freely.
4. Matplotlib is mostly written in python, a few segments are written in C, Objective-C and Javascript for Platform compatibility.

- Matplotlib is one of the most popular Python packages used for data visualization. It is a cross-platform library for making 2D plots from data in arrays.

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Installation

- use following command to install Matplotlib

  pip install Matplotlib

- To check matplotlib version
  import matplotlib
  matplotlib.__version__

If you are using Anaconda

- Start Navigator.
- In Environments tab checked for installed packages.

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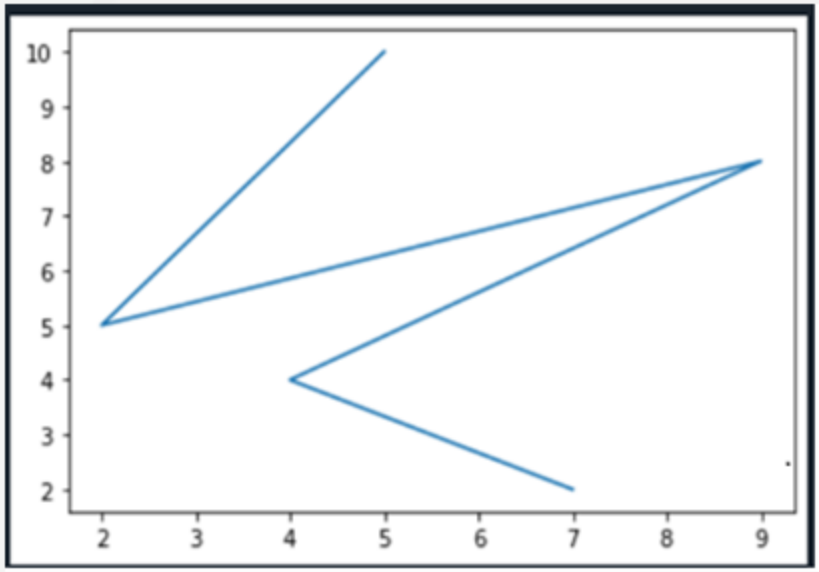
Types Of Plots

- Line Plot
- Barplot
- Scatter plot

Line Plot

- Line Plot

```
import matplotlib.pyplot as plt
x=[5,2,9,4,7]
y=[10,5,8,4,2]
plt.plot(x,y)
plt.show()
```



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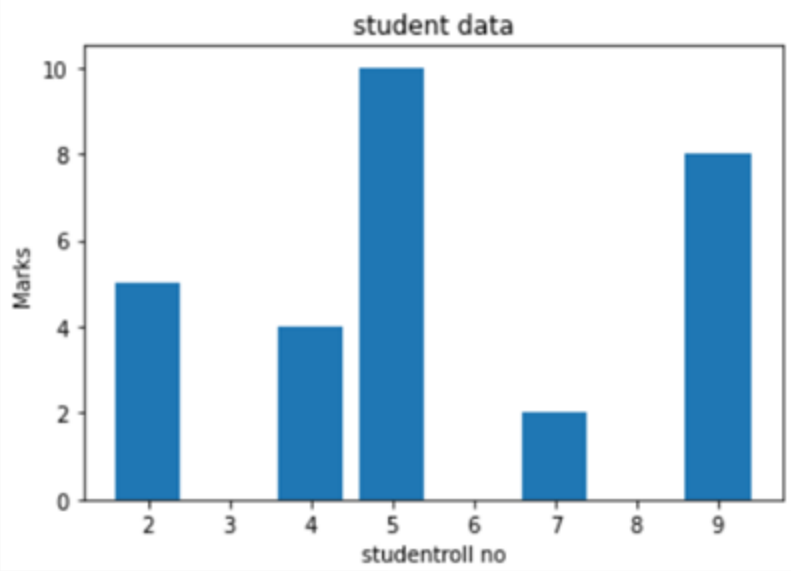
Barplot

```
import matplotlib.pyplot as plt
x=[5,2,9,4,7]
y=[10,5,8,4,2]
plt.bar(x,y)
plt.title("student data")
plt.xlabel("studentroll no")
plt.ylabel("Marks")
plt.show()
```



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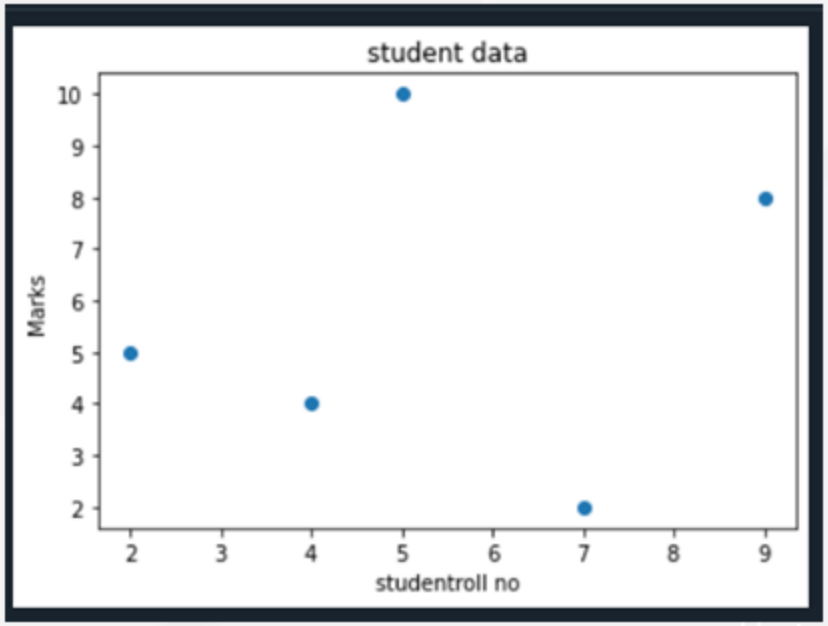
Scatter plot

```
import matplotlib.pyplot as plt
x=[5,2,9,4,7]
y=[10,5,8,4,2]
plt.scatter(x,y)
plt.title("student data")
plt.xlabel("studentroll no")
plt.ylabel("Marks")
plt.show()
```



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Thank You